

Sebastian Cruz

Orlando, FL | scruzloaiza@middlebury.edu | (689) 254-8886 | GitHub | sebastiancruz.netlify.app

EDUCATION

Middlebury College <i>Bachelor of Arts in Computer Science. College Scholar's List (Highest Academic Honors), QuestBridge Scholar. GPA: 3.82/4.00.</i> Relevant Coursework: Software Development, AI, Algorithms, Machine Learning	Middlebury, VT <i>Expected May 2027</i>
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EXPERIENCE

AI Lead and Backend Developer <i>Altheros Capital</i> - Led backend AI team as technical lead, coordinating development efforts, reviewing pull requests, assigning tasks, and contributing to Agile sprint planning and architectural decisions. - Designed and implemented Retrieval-Augmented Generation (RAG) infrastructure in Python, building document ingestion pipelines that process, chunk, embed, and store knowledge in PostgreSQL/pgvector to support AI-powered knowledge retrieval. - Built RESTful backend APIs, data models, and AI services using Python and Pydantic, researching and evaluating new technologies to improve system scalability, maintainability, and clean architecture.	Spring 2026 – Present <i>Remote</i>
Backend Developer <i>Text Maddie</i> - Developed backend workflows for AI-powered lead conversion agents, implementing Python logic that automated follow-ups, appointment scheduling, CRM updates, and integrations across HubSpot, Slack, and messaging platforms. - Engineered AI agent behaviors, workflows, and tool integrations to support natural customer conversations while triggering automated business processes and coordinating multi-agent interactions. - Diagnosed and resolved production issues through log analysis, workflow tracing, and backend debugging to ensure reliable AI agent performance.	Summer 2026 – Summer 2026 <i>Remote</i>
Research Assistant <i>Middlebury College</i> - Implemented InfoNCE and RINCE loss functions for NPSV-deep, a machine learning project for structural variant genotyping using deep neural networks in PyTorch, improving embedding-based classification accuracy by 0.5-2.9% across 4 datasets and increasing non-reference concordance by up to 3%. - Trained and evaluated deep learning models in PyTorch Lightning using TensorBoard, optimizing model parameters and maintaining working iterations via Git version control.	Summer 2025 – Summer 2025 <i>Middlebury, VT</i>

PROJECTS

MiddBin https://middbin.netlify.app <i>Scrum Master and Developer</i> - Built a full-stack marketplace web application using Next.js, React, TypeScript, Java, and Supabase (PostgreSQL), collaborating with a team through GitHub pull requests in an Agile workflow while serving as Scrum Master. - Designed and implemented a dynamic listing creation form with conditional inputs and polished UI transitions using CSS to improve usability and data validation. - Engineered a scalable image upload system using Supabase Storage, including multi-image previews, gallery views, and automatic cleanup on listing deletion, integrated with listing grids and detail pages via RESTful API calls.	Fall 2025 – Fall 2025
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SKILLS

Languages: Python, JavaScript, TypeScript, Java
Frameworks and Libraries: FastAPI, Pydantic, React, Next.js, PyTorch
Tools and Platforms: PostgreSQL, RESTful APIs, Git/GitHub, AI/ML Integration (RAG, LLM Agents), Agile/Scrum
Spoken Languages: English (Fluent), Spanish (Native), Portuguese (Intermediate)

CERTIFICATIONS

- CodePath Web Development (Intro and Intermediate)
- Google Project Management